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Wildfire smoke impacts respiratory health more than fine particles from other sources: observational evidence from Southern California

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Wildfires are becoming more frequent and destructive in a changing climate. Fine particulate matter, $PM_{2.5}$, in wildfire smoke adversely impacts human health. Recent toxicological studies suggest that wildfire particulate matter may be more toxic than equal doses of ambient $PM_{2.5}$. Air quality regulations however assume that the toxicity of $PM_{2.5}$ does not vary across different sources of emission. Assessing whether $PM_{2.5}$ from wildfires is more or less harmful than $PM_{2.5}$ from other sources is a pressing public health concern. Here, we isolate the wildfire-specific $PM_{2.5}$ using a series of statistical approaches and exposure definitions. We found increases in respiratory hospitalizations ranging from 1.3 to up to 10% with a $10 \mu g m^{-3}$ increase in wildfire-specific $PM_{2.5}$, compared to 0.67 to 1.3% associated with non-wildfire $PM_{2.5}$. Our conclusions point to the need for air quality policies to consider the variability in $PM_{2.5}$ impacts on human health according to the sources of emission.

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Fine particulate matter, i.e., particles with aerodynamic diameter $\leq 2.5 \mu\text{m}$ ($\text{PM}_{2.5}$), is the main component of wildfire smoke¹ that impacts public health^{2–5}. $\text{PM}_{2.5}$ can be inhaled into the deepest recesses of the lungs⁶ and may enter the bloodstream impairing vital organs including the lungs⁷. $\text{PM}_{2.5}$ in the United States has decreased in past decades due to environmental regulations^{5,8}, with the exception of wildfire-prone areas⁵. Wildfire $\text{PM}_{2.5}$ in the US is projected to increase with climate change along with the associated burden on human health⁹. Levels of wildfire $\text{PM}_{2.5}$ can greatly exceed those of ambient $\text{PM}_{2.5}$, spiking episodically within a short period of time (e.g., hours after the onset of a wildfire), and such high exposure levels may generate important health impacts. Current air quality standards specific to $\text{PM}_{2.5}$ from the Clean Air Act Amendments do not distinguish the sources of emission or chemical composition, implicitly considering $\text{PM}_{2.5}$ from wildfires and from other sources (e.g., ports, industrial plants, and traffic emissions) to be equally harmful to human health. This is also true in other regions of the world, as in the WHO Air Quality Guidelines (AQG)¹⁰ for example.

Though the differential toxicity of wildfire $\text{PM}_{2.5}$ as compared to other ambient sources of $\text{PM}_{2.5}$ is not well understood^{11–13}, recent animal toxicological studies suggest that particulate matter from wildfires is more toxic than equal doses from other sources such as ambient pollution^{14,15}. In vitro and in vivo studies have shown that mechanisms that may explain wildfire-specific PM higher toxicity include inflammation, oxidative stress¹⁵, or increased respiratory infection by altering pulmonary macrophages activity¹⁶. Wildfire particulate matter is mostly carbonaceous (with 5–20% elemental carbon and at least 50% organic carbon^{17,18}) and has more oxidative potential than ambient urban particulate due to the presence of more polar organic compounds¹⁹. All the above compounds in wildfire smoke tend to generate more free radicals and thus have a greater potential to cause inflammation and oxidative stress in the lung than urban ambient particulate from the same region^{20,21}. It is therefore imperative to differentiate between smoke and non-smoke $\text{PM}_{2.5}$ when assessing impacts on public health.

In epidemiological studies, it has been shown that $\text{PM}_{2.5}$ from wildfire smoke can exacerbate a range of health problems including respiratory and cardiovascular issues^{4,22,23} (although some uncertainty exists^{23,24}). Yet, to date, no study has assessed the public health impact of wildfire-specific $\text{PM}_{2.5}$ as it differs from $\text{PM}_{2.5}$ from other sources at a fine spatial resolution (e.g., zip code) and spanning multiple wildfires over a 14-year period. Previous studies assessing wildfire smoke effects on health have often been restricted to single wildfire events due to limitations in the estimation of human exposure to wildfire-specific $\text{PM}_{2.5}$, which typically relies on computationally demanding dynamical chemical transport models (CTMs). The most extensive study²⁵ to date assessed the impacts of smoke exposure in the elderly population (≥ 65 years) within the Western US during a 6-year period, but was resolved at a coarser level (county) and relied in part on CTMs for quantifying wildfire-specific $\text{PM}_{2.5}$. We compare four statistical approaches to isolate wildfire-specific $\text{PM}_{2.5}$ from other sources. These approaches do not rely on heavy computing efforts and offer the advantage of modeling daily, zip code-level wildfire-specific $\text{PM}_{2.5}$ over a long study period and extensive area.

In Southern California (SoCal), the dry gusty offshore (northeasterly) Santa Ana winds (SAW) start-up in the fall, peak in December, and wane in the spring²⁶. SAWs are thus episodic reversals of the prevailing onshore (westerly) winds in SoCal. Early season SAWs of autumn, occurring after the long dry Mediterranean summer and before the first rains of winter, typically drive the largest wildfires, while most ignitions are

human caused²⁷. The Southern California traditional wildfire season thus differs from that in most other Western US regions due to its meteorological, climatic, and ignition causes^{27,28}. We note, however, that recent winter and spring SAW-driven wildfires and research suggest that the SoCal wildfire season may be expanding²⁹. We also note that SAWs are dormant in summer, and although summer heat-driven wildfires are becoming more prevalent lately, smoke from such fires (with the notable exception of hundreds of California wildfires burning in August 2020!) does not typically impact the coastal zone, which is the reason summer is excluded from this analysis. Dry gusty SAWs accelerate and warm on their way down coastal topography towards sea level; they not only fan and spread wildfires burning in the wildland–urban interface, but also transport smoke to densely populated coastal areas. These SAW-driven wildfires can spread faster and burn longer than fires at other times of the year³⁰. Furthermore, recent work has shown that $\text{PM}_{2.5}$ tends to increase with strong SAWs in the presence of wildfire burning upwind³¹.

In this paper, we assess the impacts on respiratory health outcomes of $\text{PM}_{2.5}$ attributable to wildfire smoke in SoCal, as compared to $\text{PM}_{2.5}$ from other sources. In this region, sources of non-smoke $\text{PM}_{2.5}$, include vehicular emissions, secondary aerosols (e.g., sulfate and nitrate), soil, agricultural, and industrial emissions^{32–34}. In order to robustly isolate the health impacts of wildfire-specific $\text{PM}_{2.5}$, we apply and compare four analytical approaches. Specifically, we implemented the following distinct methods: (i) an instrumental variable approach with a two-stage regression; (ii) a spatio-temporal multiple imputation approach, and (iii) an interaction effect approach. Lastly, we also compared these approaches to (iv) a seasonal interpolation method recently proposed by Lipner et al.³⁵.

In each of these approaches, we relied on two distinct exposure variables yielding eight estimates of the differential impacts of wildfire-specific $\text{PM}_{2.5}$. We used (i) the occurrence of strong SAWs and the presence of fire upwind and (ii) smoke plume datasets (NOAA Hazard Mapping System (HMS)) within a 160 km buffer from a wildfire perimeter to identify zip code days exposed to smoke. In addition to isolating the effect of wildfire-specific $\text{PM}_{2.5}$ from other sources on respiratory health, the analytical methodology implemented in this study allowed us to cover a large region, population, and study period spanning 14 years (1999–2012).

Our study comprised 696 zip code polygons within the Santa Ana wind domain (Fig. 1). The Santa Ana wind season extends liberally from September to May²⁶ and therefore the summer months of June, July, and August were excluded from our analyses. Daily-, zip code-specific concentrations of $\text{PM}_{2.5}$ ³¹ represent fine particulate matter from all sources, including ambient levels and wildfire smoke. Daily hospital admissions for respiratory diseases ($n = 1,655,011$), which include pulmonary diagnoses such as asthma, chronic obstructive pulmonary disease (COPD), pneumonia, and interstitial lung disease were aggregated at the daily level by zip code.

The highest mean $\text{PM}_{2.5}$ concentrations were observed in highly populated coastal zip codes, as well as in some inland zip codes in San Bernardino and Riverside Counties (Fig. 2a). Mean values for rates of respiratory admissions per 100,000 individuals suggested pockets of higher incidence in some urban areas and possibly heightened admissions in the Central Valley where dust may be a factor (Fig. 2b). Total monthly regional SAW activity, i.e., the sum of the Santa Ana Wind Regional Index (SAWRI)²⁶ over a month within 1999–2012, reflecting both intensity and frequency of SAW, peaked between the months of November and January. Similarly, mean $\text{PM}_{2.5}$ values were highest during late fall and winter months, whereas peaks in respiratory admissions were observed mainly in winter and particularly during February

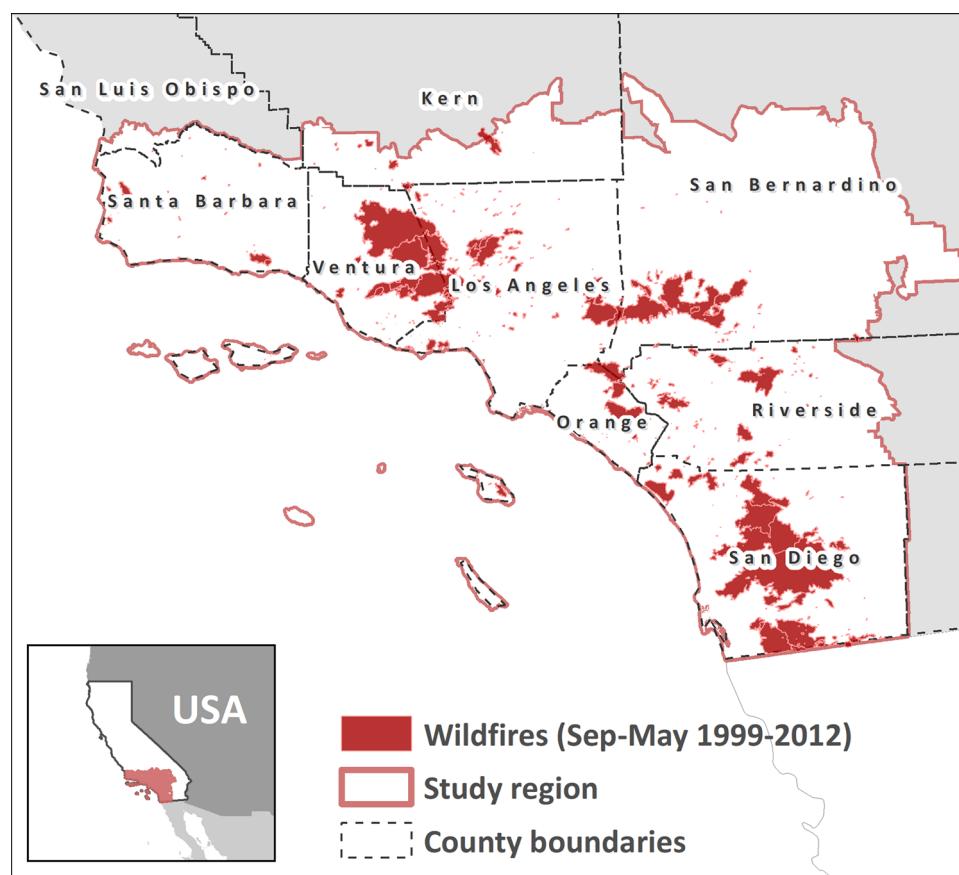


Fig. 1 Wildfire perimeters in Southern California (1999–2012). Wildfire perimeters shown here represent the total area burned for a given fire during our study period comprising 1999–2012, excluding summer months (June, July, and August). The inset figure shows the location of our study region, which includes zip codes within the Santa Ana Wind domain in Southern California, USA.

(Fig. S1 in Supplementary information). Figure 3 shows mean values for wildfire-specific $PM_{2.5}$ estimated by imputation and seasonal interpolation, using the fire upwind and strong SAW exposure definition. In terms of the approaches used to isolate wildfire-specific $PM_{2.5}$, when comparing the imputation and seasonal interpolation estimates, we find that the latter might yield larger values in some instances since the non-smoke background considers a seasonal median and thus can be lower than the non-smoke concentration imputed daily at a given zip code by means of the spatio-temporal imputation approach. In addition, we include case studies related to wildfire events that took place in October 2007, the most impactful in terms of wildfire smoke exposure and burden to public health³⁶ (see details and resulting figure in Supplementary information). This case studies further illustrate that wildfire-specific $PM_{2.5}$ estimates widely agree during extreme wildfire events such as the aforementioned 2007 firestorm.

Results and discussion

Effects of wildfire- $PM_{2.5}$ on respiratory admissions. Table 1 summarizes our results for the effects of wildfire- $PM_{2.5}$ on respiratory admissions in Southern California over the period 1999–2012, excluding summer months (June, July, and August), using the occurrence of strong SAWs and the presence of fire upwind as exposure definition. Based on the mean number (1.85) of daily respiratory admissions per 100,000 individuals, a $10 \mu g m^{-3}$ increase in $PM_{2.5}$ was estimated to increase the number of admissions by only 0.76% (95% CI: 0.42–1.1). In contrast, the causal effects of $PM_{2.5}$ attributable to wildfire smoke estimated by

spatio-temporal imputation amounted to a 10.0% (95% CI: 3.5–16.5) increase in admissions, the highest percentage among all methods and exposures used. Such results were similar, though varying in the amplitude of % increase in admissions when using other approaches to isolate the wildfire-specific $PM_{2.5}$ (see Table 1), as well as among all approaches considering smoke plumes within a 160 km buffer to define zip code days exposed to wildfire smoke (see Table S1 in Supplementary information). We conclude that wildfire-specific $PM_{2.5}$ is up to 10 times more harmful on human health than $PM_{2.5}$ from other sources. All the above methods and resulting estimates in increased admissions have, however, very wide confidence intervals, but these estimates are consistently higher than their aggregated or non-smoker counterparts.

Study limitations include the use of patient home address to estimate exposures and using community-level $PM_{2.5}$ to assess and quantify individual wildfire $PM_{2.5}$ exposures. The number and extent of smoke plumes used to categorize exposed zip code days represent a conservative estimate due to the limitations of visible satellite data. In addition to all the above, our definition of upwind fire exposure (detailed in “Methods”) may have also misclassified some of the smoke $PM_{2.5}$ as non-smoke $PM_{2.5}$ and vice versa. The fire upwind and strong SAW exposure definition focuses exclusively on SAW-driven wildfires and the overall north-easterly wind direction. The smoke plumes and buffer exposure definition, on the other hand, may include a few small non-SAW wildfires that occurred during the September–May period in a given year. Smoke from such inland wildfires, however, tends to be transported away from the coast by the prevailing onshore winds. This is also the case with summer wildfires. Although wildfires are burning in August 2020, at the

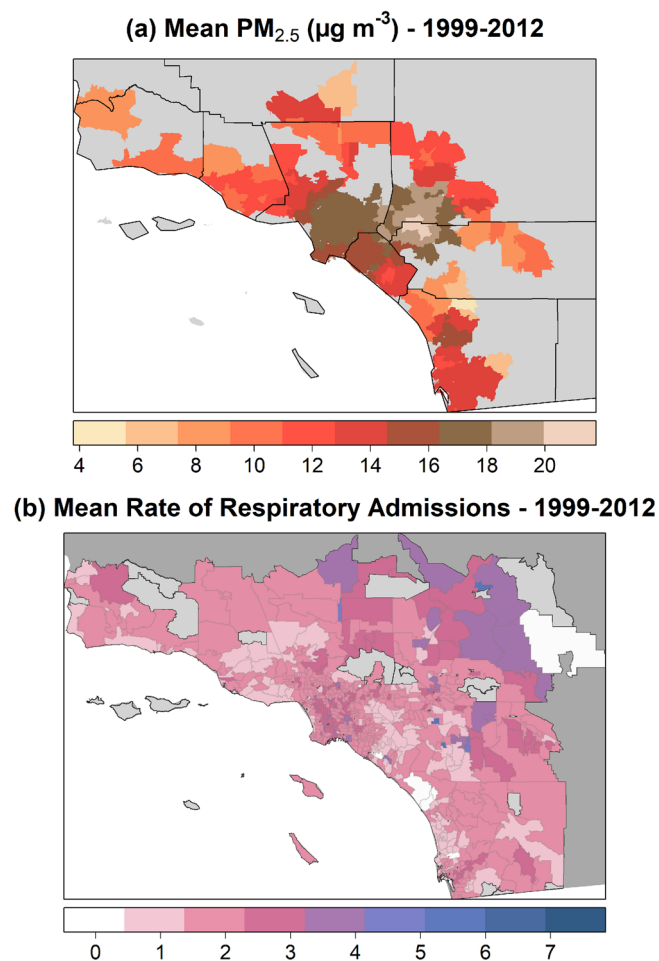


Fig. 2 Mean values of $\text{PM}_{2.5}$ and rate of respiratory admissions by ZIP code. **a** Mean $\text{PM}_{2.5}$ concentrations at available zip codes (county boundaries shown in black) and **b** mean rate of respiratory admissions (per 100,000 individuals) per zip code during 1999–2012 (summer months not considered).

time of this writing, Santa Ana winds are dormant in summer, and smoke from such wildfires does not typically impact the coastal zone. This is the reason summer was excluded from our analysis.

Additional potential limitations involve not including any lagged effects in our models when assessing the impact of exposure to smoke $\text{PM}_{2.5}$ on respiratory health. Lastly, we acknowledge that wildfires can increase tropospheric ozone, which is a powerful oxidant that can irritate the airways and can thus increase the risk of hospitalizations for respiratory conditions³⁷. At the same time, it has been shown that wildfires generate increases in ozone levels through processes distinct from $\text{PM}_{2.5}$ from smoke^{38,39}. Furthermore, a recent paper showed that $\text{PM}_{2.5}$ was associated with respiratory ED visits and hospitalizations during a wildfire period even when adjusted for ozone⁴⁰. Recent studies proposed different methods to predict ozone exposure during wildfire events⁴¹ and future studies should address isolating the health impacts of ozone generated specifically by wildfires.

Implications for public health and air quality policy. Our findings indicate that wildfire-specific $\text{PM}_{2.5}$ can cause a greater impact on respiratory health than $\text{PM}_{2.5}$ from other sources. In each approach and each combination of variables to define zip code days exposed to wildfire plumes, we found that wildfire-specific $\text{PM}_{2.5}$ were up to 10 times more harmful than non-smoke $\text{PM}_{2.5}$. Wildfires have the potential to greatly and suddenly increase $\text{PM}_{2.5}$

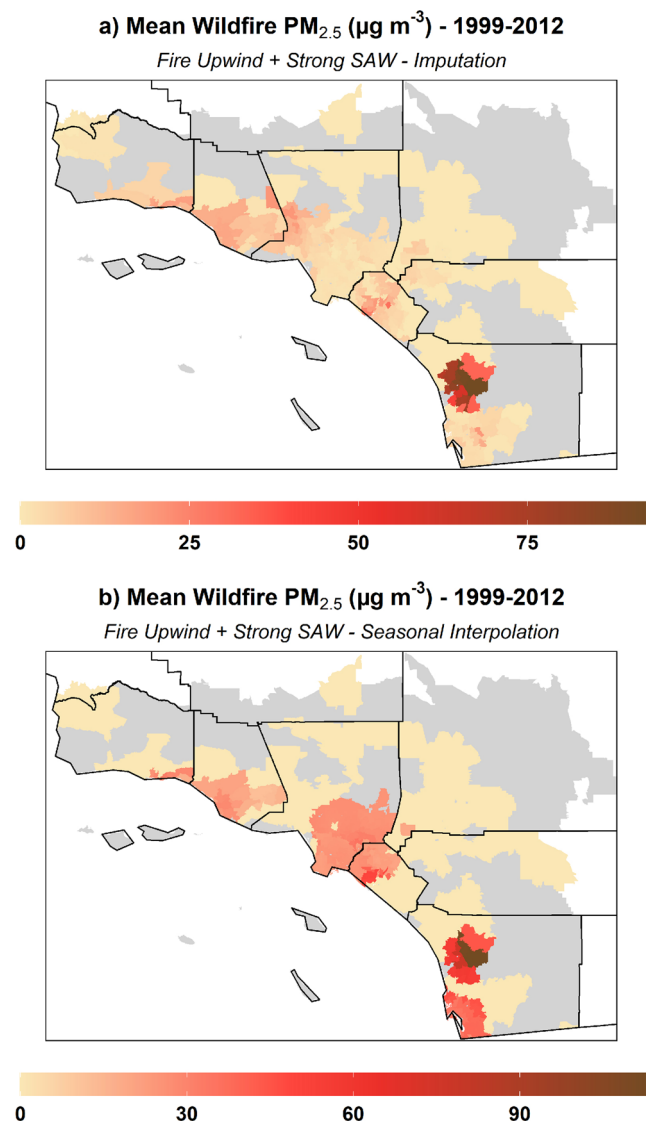


Fig. 3 Wildfire-specific concentrations of $\text{PM}_{2.5}$. Mean wildfire-specific $\text{PM}_{2.5}$ estimated by **a** imputation and **b** seasonal interpolation methods and using the fire upwind and strong SAW exposure definition.

concentrations^{22,36}, often surpassing safe limits ($35 \mu\text{g m}^{-3}$) and reaching levels qualified as hazardous ($>250 \mu\text{g m}^{-3}$) by the Air Quality Index (AQI, US EPA). Such sudden increase in $\text{PM}_{2.5}$ caused by wildfire smoke can thus particularly affect vulnerable populations such as children and the elderly^{23,36,42,43}. Overall, a greater impact of wildfire smoke $\text{PM}_{2.5}$ on public health relative to ambient levels can be expected as $\text{PM}_{2.5}$ concentration tends to be higher during wildfire episodes. However, in this study, we also show that even for similar exposure levels, $\text{PM}_{2.5}$ from wildfires is considerably more dangerous for respiratory health. A comparable study²⁵ examining the elderly population in counties across the Western US found a 7.2% increase in the risk of respiratory admissions during smoke days with high wildfire-specific $\text{PM}_{2.5}$ ($>37 \mu\text{g/m}$) compared with nonsmoker days.

Recent toxicological studies have shown differences in the composition and effects of wildfire $\text{PM}_{2.5}$ compared to ambient sources^{14,15,44,45}. In one study¹⁴, significant changes were observed in macrophage and neutrophil counts in mouse lung samples exposed to wildfire particulate matter compared to ambient sources. Specifically, the authors observed that the toxicity of PM in wildfire smoke to the respiratory system is 3–4